OFFICIAL BIDDING DOCUMENT

**Telecommunications Technology Solutions Inc. (TelTech)**

Corporate Tender Secretariat · Procurement & Network Engineering Division

**RFP-TELTECH-FWA-2026-001 / APPENDIX A**
**Issuing Body:** Technical Evaluation Board
**Date:** October 14, 2026

# Technical Specifications & Compliance Matrix

Mandatory Technical Requirements for RAN, Power, 5GC, OSS/BSS, RF, Managed Services & CPE

| DOCUMENT NUMBER | ISSUING BODY | TARGET SPECTRUM | BIDDING STATUS |
|---|---|---|---|
| RFP-TELTECH-FWA-2026-001 | Technical Evaluation Board | Band n50 (1.5 GHz, 60 MHz TDD) | Official Request for Proposal |

*Instructions to Bidders: Complete Column D with "C" (Full Compliance), "M" (Minor Deviation - explain in remarks), or "NC" (Non-Compliant). Bidders must provide specific technical values and document references.*

## SECTION 1: RADIO ACCESS NETWORK (RAN) TECHNICAL SPECIFICATIONS

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| RAN-01 | 3GPP Standard | 3GPP Release 16 and Release 17 compliant 5G NR Standalone (SA) architecture. | | |
| RAN-02 | Frequency Band | Band n50 (1432 – 1517 MHz, 1.5 GHz L-Band TDD). Full RF support for 1432–1517 MHz. | | |
| RAN-03 | Channel Bandwidth | Continuous 60 MHz (NCR / Meralco footprint, 162 RBs) and 80 MHz (Rest of Country / Provincial footprint, 217 RBs) single carrier; software-configurable to 40 MHz / 50 MHz / 60 MHz / 80 MHz. | | |
| RAN-04 | Duplex Mode | Time Division Duplex (TDD) with flexible frame configuration (e.g., 4:1 DL:UL ratio with special slot 10:2:2). | | |
| RAN-05 | Subcarrier Spacing | 30 kHz Subcarrier Spacing (SCS) with normal cyclic prefix (CP). | | |
| RAN-06 | Synchronization | IEEE 1588v2 Precision Time Protocol (PTP G.8275.1 profile), Synchronous Ethernet (SyncE), and GNSS (GPS/BeiDou/GLONASS) receiver onboard BBU. | | |
| RAN-07 | Radio Configuration | Standard macro site: 3 sectors, 8T8R configuration (3x 8T8R RRU or integrated AAU per site). | | |
| RAN-08 | RF Output Power | Minimum 8x 40W (320W total RF output power per sector/RRU). Digital power control with dynamic power backoff. | | |
| RAN-09 | Beamforming | Digital + analog beamforming (horizontal and vertical); dynamic Multi-User MIMO (MU-MIMO) pairing in DL and UL. | | |
| RAN-10 | Modulation Schemes | Downlink: Up to 256-QAM adaptive modulation.<br>Uplink: Up to 64-QAM / 256-QAM adaptive modulation. | | |

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| RAN-11 | MIMO Support | Downlink: Up to 8-layer transmission across multiple users (MU-MIMO); up to 4-layer transmission to single 4R CPE.<br>Uplink: Up to 2-layer single user MIMO; 4-layer MU-MIMO. | | |
| RAN-12 | Sector Throughput | Aggregated single-sector peak throughput >= 450 Mbps in 60 MHz DL (>= 600 Mbps in 80 MHz DL) under 4:1 TDD frame configuration. | | |
| RAN-13 | Subscriber Capacity | BBU and sector software must support >= 350 active concurrent RRC connected CPEs per sector (>= 1,000 per 3-sector site) without degradation. | | |
| RAN-14 | BBU Architecture | Modular 19" rack-mountable subrack (<= 3U height). Dual hot-swappable -48V DC power supplies. Redundant main control units. | | |
| RAN-15 | Fronthaul Interfaces | Minimum 6x 25GE eCPRI optical interfaces per BBU; support for fiber-saving BiDi transceivers. | | |
| RAN-16 | Backhaul Interfaces | Minimum 2x 10GE + 2x 25GE SFP28 interfaces supporting IP routing, VLAN tagging (802.1Q), and Segment Routing / IP MPLS. | | |
| RAN-17 | Environmental Rating | Outdoor RRU/AAU: IP65 / IP67 ingress protection; operating temperature -40°C to +55°C; wind load resistance up to 250 km/h. | | |
| RAN-18 | Energy Saving Modes | Micro-sleep, symbol shutdown, deep sleep during low-traffic periods, reducing idle power consumption by >= 40%. | | |
| RAN-19 | Antenna Specifications | 8-port cross-pol antenna, 65° horizontal HPBW, >= 17.5 dBi gain, integrated AISG 2.0 internal remote electrical tilt (eRET) adjustable 2° to 12°. | | |
| RAN-20 | PIM Performance | Passive Intermodulation (PIM) <= -153dBc @ 2 x 43 dBm carrier test. | | |

## SECTION 2: POWER SYSTEMS & PASSIVE SITE INFRASTRUCTURE

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| PWR-01 | DC Power Plant | 19" 3U/4U modular high-efficiency switchmode rectifier system; continuous rating >= 12.0 kW total installed / 9.0 kW firm N+1 capacity @ -48V DC (4 x 3.0 kW modules: 3 active + 1 redundant standby). | | |
| PWR-02 | Rectifier Modules | Hot-swappable 3.0 kW modules in N+1 redundant configuration (4x 3.0 kW modules total, 3 active + 1 standby). Operating efficiency >= 96.0%. | | |
| PWR-03 | Input Voltage Range | Wide AC input operating range: 85V AC to 300V AC nominal single phase (230V, 60 Hz), withstanding rural grid fluctuations. | | |

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| PWR-04 | Lithium Battery (BESS) | Lithium Iron Phosphate (LiFePO4 / LFP) battery energy storage system: Sized to deliver at least 12.0 kWh usable energy to 3.0 kW DC continuous BTS load over 4.0 continuous hours without AC power, dimensioned for max 80% usable DoD and 80% end-of-life SoH retained capacity (minimum >= 20.0 kWh nominal installed capacity, minimum 400 Ah to 450 Ah @ 48V, dimensioned via mandatory loss-based engineering calculation delivering 12.0 kWh net usable energy at 80% DoD and 80% EOL SoH). Bidder shall submit formal load-autonomy calculation sheet, manufacturer discharge curves, and guarantee 4.0h autonomy at SAT. | | |
| PWR-05 | Battery Autonomy | Guaranteed >= 4.0 hours continuous operation at 3.0 kW continuous BTS load without AC utility input, verified by full discharge test at site acceptance. | | |
| PWR-06 | Battery Cycle Life | >= 4,000 cycles at 80% Depth of Discharge (DoD) at +25°C; built-in intelligent Battery Management System (BMS). | | |
| PWR-07 | Standby Genset | For Greenfield (Class T4) sites: 15 kVA / 12 kW prime, 230V, 60 Hz diesel generator in sound-attenuated weatherproof canopy (<68dBA @ 7m). | | |
| PWR-08 | Fuel Autonomy | Integrated sub-base fuel tank with minimum 500L–1,000L capacity providing >= 72hours continuous runtime under full load. | | |
| PWR-09 | Automatic Transfer (ATS) | Microprocessor-controlled ATS with 4-pole motorized breaker and mechanical interlock, transferring power within 30 seconds of grid failure. | | |
| PWR-10 | Surge Protection | Coordinated SPD Type 1+2 on AC input (>= 40kA) and DC distribution busbar (>= 20kA). | | |
| PWR-11 | TowerCo Tap-Off Kit | Standardized co-location DC tap box with integrated Class 1.0 revenue-grade kWh energy meter and RS485/Modbus telemetry. | | |
| PWR-12 | Grounding & Lightning | Comprehensive grounding system achieving <= 5.0 Ohms earth electrode resistance; copper busbars and copper-clad steel ground rods. | | |

## SECTION 3: 5G CORE NETWORK (5GC SA) & DATACENTER INFRASTRUCTURE

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| 5GC-01 | Architecture | Cloud-native, containerized microservices 5G Standalone (5GC SA) compliant with 3GPP Release 16/17 Service-Based Architecture (SBA). | | |
| 5GC-02 | Network Functions | Full NF suite: AMF, SMF, UPF, PCF, UDM, UDR, AUSF, NRF, NSSF, NEF, CHF, and SEPP. | | |
| 5GC-03 | Subscriber Sizing | Day-1 licensed capacity: 1,000,000 registered FWA subscribers. Scalable in 500k increments up to 3,000,000 subscribers. | | |
| 5GC-04 | Geo-Redundancy | Active-Active geo-redundant dual-datacenter deployment (Metro Manila Primary DC and Clark Pampanga DR DC). | | |
| 5GC-05 | Session-Preserving Failover | Automatic subscriber session continuity (zero session loss across UDR/UDM and SMF clusters) and automated user-plane traffic restoration <= 30 seconds during datacenter failure or scheduled DR drill. | | |

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| 5GC-06 | **UPF Throughput** | UPF dimensioned for >= 80 Gbps usable wire-speed throughput per 2U dual-socket compute node with DPDK and SR-IOV acceleration. | | |
| 5GC-07 | **Carrier-Grade NAT** | Integrated or inline CG-NAT supporting deterministic port allocation, logging per NTC legal intercept mandate, and 10:1 to 16:1 subscriber ratio. | | |
| 5GC-08 | **Deep Packet Inspection** | Embedded DPI capable of Layer-7 application recognition, speed throttling per subscriber tariff tier, and fair-use policy enforcement. | | |
| 5GC-09 | **QoS & Slicing** | 5G QoS Flow mapping (5QI 1, 2, 7, 8, 9) and Network Slicing (S-NSSAI) separating consumer FWA, SME enterprise, and management traffic. | | |
| 5GC-10 | **Datacenter Compute** | Enterprise 2U server nodes (Dual Intel Xeon Gold / AMD EPYC, 512GB RAM, redundant NVMe enterprise SSDs, dual 100GE NICs). | | |
| 5GC-11 | **Datacenter Fabric** | Leaf-spine datacenter networking architecture (32 x 100GE spine switches, 48 x 25GE + 6 x 100GE leaf switches) with EVPN-VXLAN. | | |
| 5GC-12 | **Security & Scrubbing** | Redundant carrier-grade perimeter next-generation firewalls (>= 100 Gbps throughput) and inline anti-DDoS scrubbing appliances. | | |

## SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| BSS-01 | **Product Catalog & Tariffs** | Configurable telecom product catalog supporting 4 consumer FWA plans (₱599, ₱799, ₱999, ₱1,299), SME tiers, and flexible speed-cap policies. | | |
| BSS-02 | **Prepaid Charging (CHF)** | Real-time convergent charging system (CCS/CHF) integrated via 3GPP Nchf/Diameter Ro, supporting real-time balance rating and session cutoff. | | |
| BSS-03 | **Omnichannel Payments** | Native REST API integration with Philippine payment gateways: GCash, Maya (PayMaya), 7-Eleven CLiQQ OTC, Bayad Center, and QR Ph. | | |
| BSS-04 | **Subscriber Portal & App** | Native Android/iOS subscriber self-care mobile app and responsive web portal for balance checking, online top-up, and speed-test diagnostics. | | |
| OSS-01 | **Field Force App (FMS)** | Mobile app for field technicians: Real-time work-order dispatch, turn-by-turn navigation, barcode/QR scan of CPE serials, and digital e-signature. | | |
| OSS-02 | **RF Signal Verification** | Mobile field tool must perform live outdoor RF signal validation (RSRP >= -95 dBm, SINR >= 10 dB) before allowing install closure. | | |
| OSS-03 | **Dispatch Throughput** | Algorithmic dispatch engine capable of dynamically optimizing and dispatching >= 3,500 installation and maintenance work orders daily. | | |
| OSS-04 | **CPE Inventory & WMS** | Serialized lifecycle tracking of CPE hardware across central warehouse, technician vehicle float, active subscriber homes, and repossession. | | |

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| OSS-05 | Reverse Logistics | Reverse logistics workflow tracking recovered churned CPEs through testing, factory reset, cleaning, re-boxing, and return to stock (>= 70% target). | | |
| OSS-06 | Fault & Performance OSS | Unified multi-vendor EMS/NMS monitoring 5,000 RAN sites, transmission links, and 5G Core with automated RCA and alarm correlation. | | |
| OSS-07 | Regulatory Compliance | Automated generation of quarterly NTC Quality of Service (QoS) and broadband speed performance audit reports. | | |

## SECTION 5: RF PLANNING, SITE SURVEY & CLUSTER OPTIMIZATION

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| RFP-01 | Planning Tool | Carrier-grade RF planning tool (Atoll, Asset, Planet) with high-resolution digital terrain and clutter data (<= 5m resolution). | | |
| RFP-02 | Propagation Model | Calibrated Band n50 (1.5 GHz) 3D ray-tracing / standard propagation model calibrated against local Philippine terrain clutter. | | |
| RFP-03 | Nominal Siting & Audit | Siting validation against 13,810 TowerCo registry inventory (13,799 unique IDs / 13,629 coordinates) and Converge fiber network backbone routes (subject to verification on the ground); production of candidate search rings. | | |
| RFP-04 | Technical Site Survey | Comprehensive TSSR per site including tower structural load audit, wind-load calculation, panoramic drone photography (360°), and power audit. | | |
| RFP-05 | Coverage Target | Guaranteed outdoor RSRP >= -95 dBm and SINR >= 10 dB across >= 90% of residential settlement footprint in target barangay. | | |
| RFP-06 | Single Site Acceptance | Single Site Verification (SSV) stationary and drive tests verifying sector radiation, antenna RET, PCI, TAC, and max throughput. | | |
| RFP-07 | Cluster Tuning | Cluster optimization across 10–20 site clusters: Interference analysis, tilt/azimuth physical optimization, handover parameter tuning. | | |

## SECTION 6: MANAGED SERVICES & 24/7 NETWORK OPERATIONS (MS)

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| MS-01 | 24/7 Tier-2/Tier-3 NOC | Continuous 24/7/365 active monitoring, incident triage, alarm dispatch, and escalation management across all network domains. | | |
| MS-02 | Network Availability | Overall end-to-end radio and core network service availability >= 99.70% monthly SLA. | | |
| MS-03 | Critical Incident MTTR | Mean Time to Restore full service for Critical / Severity-1 outages: <= 2hours in urban areas; <= 4hours in rural areas. | | |

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| MS-04 | First Line Maint. (FLM) | Dedicated nationwide field dispatch teams with dedicated 4x4 vehicles, fusion splicers, OTDRs, spectrum analyzers, and climbing gear. | | |
| MS-05 | Preventive Maintenance | Quarterly scheduled site inspections: Antenna bracket torque checks, RF cable weatherproofing, battery impedance tests, genset runs. | | |
| MS-06 | Spare Parts Logistics | Automated SPMS maintaining regional buffer stock; guaranteed Next Business Day (NBD) hardware replacement delivery nationwide. | | |

## SECTION 7: 5G CUSTOMER PREMISES EQUIPMENT (CPE - LOT 7)

| ITEM ID | SUB-SYSTEM / FEATURE | DETAILED SPECIFICATION REQUIREMENT | COMPLIANCE (C / M / NC) | BIDDER PROPOSED SPECIFICATION & REFERENCE |
|---|---|---|---|---|
| CPE-01 | 3GPP Standards & Modes | 3GPP Release 16/17 compliant 5G NR sub-6 GHz Standalone (SA) and Non-Standalone (NSA) operation. | | |
| CPE-02 | Frequency Bands | Native Band n50 (1432 – 1517 MHz, supporting 60 MHz and 80 MHz channel bandwidths) plus fallback sub-6 GHz bands (n1, n3, n7, n28, n41, n77, n78). | | |
| CPE-03 | Indoor CPE Antenna & MIMO | High-gain omnidirectional multi-element internal antenna array (minimum >= 8.0 dBi gain across Band n50); 4x4 MIMO DL, 2x2 MIMO UL. | | |
| CPE-04 | Outdoor CPE Antenna & MIMO | High-gain directional integrated antenna array (minimum >= 12.0 dBi gain across Band n50); 4x4 MIMO DL, 2x2 MIMO UL; adjustable pole/wall mounting bracket. | | |
| CPE-05 | Wi-Fi Subsystem | Dual-band Wi-Fi 6 (802.11ax) 2.4 GHz (2x2) and 5 GHz (2x2) concurrent, OFDMA, MU-MIMO, WPA3-Personal encryption. | | |
| CPE-06 | Ethernet Interfaces | Indoor CPE: Minimum 1x 2.5GE + 2x 1GE RJ-45 LAN ports. Outdoor CPE: 1x 2.5GE PoE input port plus companion indoor Wi-Fi 6 router/gateway and 802.3at PoE injector included in kit. | | |
| CPE-07 | Environmental Rating | Outdoor CPE: IP67 dust/water ingress protection, operating temperature -10°C to +55°C, 200 km/h wind survivability. Indoor CPE: IP20, 0°C to +40°C. | | |
| CPE-08 | Remote Device Management | Broadband Forum TR-069 and TR-369 (USP) support with complete TR-181 data model, SNMPv2/v3, and secure ACS provisioning. | | |
| CPE-09 | Firmware & Security | Secure Boot with cryptographic signature verification, automated FOTA (Firmware-Over-The-Air) batch upgrades, and operator lock capability. | | |
| CPE-10 | SIM / eSIM Support | Standard Nano-SIM (4FF) slot with anti-theft locking cover and optional GSMA-compliant eSIM MFF2 footprint. | | |
| CPE-11 | Power Supply | Indoor CPE: 230V AC / 12V DC external adapter (Philippine standard Type A/B plug). Outdoor CPE: 2.5GE-capable 802.3at/bt PoE power injector and 2.5GE WAN indoor gateway companion included. | | |
| CPE-12 | Certification & Warranty | NTC Type Approval compliance, CE/FCC certification, and minimum 24-month hardware replacement warranty (RMA <= 5 business days). | | |

*(End of Appendix A — Technical Specifications & Compliance Matrix)*